SYNAPSE EDGE DEFENSE

# Campaign Correlation Engine

Seven weighted detectors automatically link related attacks into unified campaigns — the work SOC analysts do manually, done in microseconds at the edge.

| 7 | <50µs | 100% | 24hr |
|---|---|---|---|
| DETECTION ALGORITHMS | CORRELATION LATENCY | LOCAL — ZERO CLOUD | SIGNAL RETENTION |

## 7 CORRELATION DETECTORS

Each detector runs independently, producing a weighted confidence score

**50 — Attack Sequence**
Same attack payloads appearing across different source IPs.
CATCHES: Coordinated vulnerability scanning, distributed SQLi/XSS campaigns

**45 — Auth Token**
Same JWT structure or issuer appearing across multiple IPs.
CATCHES: Credential stuffing, token replay, shared bot infrastructure

**40 — HTTP Fingerprint**
Different IPs with identical browser fingerprints (JA4H).
CATCHES: IP-rotating bots, residential proxy attacks, same tooling

**35 — TLS Fingerprint**
Same TLS signature (JA4) across different sessions.
CATCHES: Scanner frameworks, bot toolkits, headless browser farms

**30 — Behavioral Similarity**
Identical navigation and timing patterns across actors.
CATCHES: Scripted attacks, automated reconnaissance, identical crawl paths

**25 — Timing Correlation**
Coordinated request timing patterns across IPs.
CATCHES: Botnets, DDoS precursors, synchronized attack waves

**15 — Network Proximity**
Same ASN or /24 subnet clustering. Lowest weight — supporting evidence, not primary signal.
CATCHES: Hosting provider abuse, VPS farm attacks, related infrastructure

## CONFIDENCE THRESHOLDS

| 96+ | 80–95 | 60–79 | 40–59 |
|---|---|---|---|
| AUTO-BLOCK | CHALLENGE | RATE LIMIT | MONITOR |

- Confirmed campaign — block all associated actors fleet-wide
- High confidence — issue progressive challenges via Interrogator
- Medium — apply rate limiting, continue monitoring signals
- Low — log signals for investigation, no enforcement action

## SIGNAL INTELLIGENCE — 4 CATEGORIES

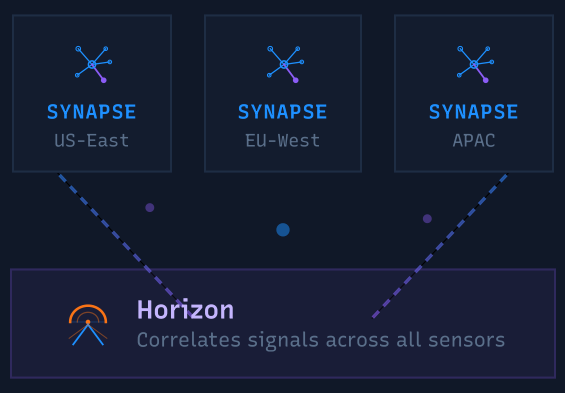
**Auth Tokens**
JWT structure
Issuer pattern
Claims analysis
Rotation timing

**Device Fingerprints**
Browser profile
Screen + plugins
Canvas hash
JA4H signature

**Network Signals**
IP + ASN
TLS fingerprint
Geolocation
Subnet clustering

**Behavioral Signals**
Navigation paths
Request timing
Sequence patterns
Session cadence

## FLEET MODE — CROSS-SENSOR CORRELATION




**Cross-Sensor Campaigns**
Attacks spanning multiple locations unified into single campaigns

**Global Block Lists**
Block propagated from one sensor to entire fleet instantly

**Unified Actor Profiles**
Single view of threat actors across your entire fleet

**War Rooms**
Real-time incident coordination with full campaign context

> **The brain is in the sensor. Campaign correlation happens at the edge.**
> No cloud latency. No backend bottleneck. Instant detection and response.

SYNAPSE · HORIZON

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Edge Protection Platform · Atlas Crew